ORACLE DATABASE 19c

Basic Administration & Practical Use Cases

A hands-on starter guide for a single-instance Windows Server lab

Version 1.1 | 3 November 2026

Environment covered

ITEM	THIS LAB
Windows host	Windows Server 2025 Standard Evaluation (build 26100)
Oracle software	Oracle Database 19c Enterprise Edition 19.3.0.0.0
Oracle Home	C:\app\oracle\product\19.0.0\dbhome_1
Container database / SID	ORCLCDB
Pluggable database	ORCLPDB1
Oracle Net listener	LISTENER on TCP 1521
Existing tablespace	APP_DATA (100 MB initial, autoextend to 1 GB)
Database logging	ARCHIVELOG enabled

CERTIFICATION Oracle 19.3 on Windows Server 2025 is a working lab configuration here, but Oracle documents a minimum 19c RU of 19.27 for Windows Server 2025. Do not treat this as a supported production build.

The steps below are training procedures, not a record of tasks already executed. Your CDB, PDB, listener, saved PDB state and APP_DATA tablespace were verified previously; APP_USER, sample data, backups and export/import remain exercises unless separately completed.

How to use this guide

- Read the command prompt label before copying anything: PowerShell, SQL> and RMAN> accept different command languages.
- Begin with read-only health checks, then create sample objects only in ORCLPDB1.
- Use new lab credentials; never put actual administrative passwords in shared documents or screenshots.
- Treat shutdown, restart, backup configuration, user creation and grants as change operations.

Quick navigation

AREA	USE CASE
Daily health	01 - Validate Oracle services, CDB, PDB and listener
PDB lifecycle	02 - Open PDB and preserve state
Schema and data	03 - Create APP_USER and populate CUSTOMERS
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Before you begin: shells, privileges and safe defaults

PROMPT	USE IT FOR	EXAMPLE
PS C:\>	Windows OS, services, file paths, networking	Get-Service *Oracle*
SQL>	SQL and SQL*Plus commands against a database	SHOW PDBS;
RMAN>	Backup, validation and recovery commands	LIST BACKUP SUMMARY;

Set a working Oracle command environment

In a new Administrator PowerShell session, execute:

```
$oh = 'C:\app\oracle\product\19.0.0\dbhome_1'
$env:ORACLE_SID = 'ORCLCDB'
& "$oh\bin\sqlplus.exe" / as sysdba
```

For OS-authenticated SQL*Plus connections you should see a SQL> prompt. To use SQL*Plus over TCP, use a database account and the PDB service:

```
-- At the SQL> prompt (not PowerShell):
CONNECT SYSTEM@"127.0.0.1:1521/orclpdb1"
-- Enter the SYSTEM password at the prompt.
```

PASSWORD NOTE Your earlier ORA-12543 incident was resolved by enclosing the administrative password containing @ in double quotes at the SQL*Plus password prompt. Do not add a password to the login command.

CDB root versus PDB: always know where you are

```
SHOW CON_NAME;
SHOW PDBS;

-- Only an administrator in CDB$ROOT should switch containers:
ALTER SESSION SET CONTAINER = ORCLPDB1;
SHOW CON_NAME;
```

SCOPE Create local application users and tables inside ORCLPDB1. Do not create sample application objects in CDB$ROOT or PDB$SEED.

USE CASE 01 | Daily health checks

Confirm Windows services, listener, CDB and PDB are available. **READ ONLY**

1. Windows: inspect Oracle services

```
Get-Service -Name 'OracleServiceORCLCDB', `
  'OracleOraDB19Home1TNSListener'
& "$oh\bin\lsnrctl.exe" status LISTENER
```

Expected: both services Running; listener reports TCP 1521 and services ORCLCDB / orclpdb1 with READY handlers.

2. SQL: verify open modes and logging

```
SELECT name, open_mode, log_mode, cdb FROM v$database;
SHOW PDBS;
SELECT instance_name, status FROM v$instance;
```

Expected: ORCLCDB READ WRITE, ARCHIVELOG, CDB=YES; ORCLPDB1 READ WRITE. The seed PDB stays READ ONLY.

USE CASE 02 | PDB start/stop administration

Check saved state, and open the PDB only when required. **CHANGES STATE**

1. Inspect state (read-only)

```
-- Run as SYSDBA in CDB$ROOT
SHOW PDBS;
SELECT con_name, state, restricted
FROM dba_pdb_saved_states;
```

2. Correct PDB state only if needed

```
-- Use only if ORCLPDB1 is mounted/closed:
ALTER PLUGGABLE DATABASE ORCLPDB1 OPEN;
ALTER PLUGGABLE DATABASE ORCLPDB1 SAVE STATE;
```

Your lab already has saved state OPEN/NO, so the commands above are usually unnecessary. Do not close the PDB during active lab sessions.

USE CASE 03 | Create sample schema, table and records

Create a local owner account and populate a small data set. **CHANGES DATA**

1. Check that APP_DATA already exists

```
ALTER SESSION SET CONTAINER = ORCLPDB1;
SELECT tablespace_name, status
FROM dba_tablespaces WHERE tablespace_name = 'APP_DATA';
SELECT username FROM dba_users WHERE username = 'APP_USER';
```

Your APP_DATA tablespace was already created and verified. If APP_USER exists, skip account creation and review its current grants instead of recreating it.

2. Create APP_USER, if absent

Run as SYSDBA within ORCLPDB1. SQL*Plus prompts for a password without echoing it. Avoid double quote characters inside this demonstration password.

```
SET VERIFY OFF
ACCEPT app_pw CHAR PROMPT 'New APP_USER password: ' HIDE
CREATE USER APP_USER IDENTIFIED BY "&app_pw"
  DEFAULT TABLESPACE APP_DATA
  TEMPORARY TABLESPACE TEMP
  QUOTA 200M ON APP_DATA
  ACCOUNT UNLOCK;
UNDEFINE app_pw
GRANT CREATE SESSION, CREATE TABLE TO APP_USER;
```

3. Connect as APP_USER to the PDB

```
CONNECT APP_USER@"127.0.0.1:1521/orclpdb1"
SHOW USER;
SHOW CON_NAME;
```

Enter APP_USER password at the hidden prompt. The expected user is APP_USER, and container ORCLPDB1.

4. Create a CUSTOMERS table and insert three records

```
CREATE TABLE customers (
  customer_id NUMBER GENERATED BY DEFAULT AS IDENTITY
    CONSTRAINT pk_customers PRIMARY KEY,
  customer_name VARCHAR2(80) NOT NULL,
  city VARCHAR2(60),
  email VARCHAR2(120) UNIQUE,
  created_at TIMESTAMP DEFAULT SYSTIMESTAMP NOT NULL
);
INSERT INTO customers (customer_name,city,email)
 VALUES ('Asha Mehta','Hyderabad','asha@example.com');
INSERT INTO customers (customer_name,city,email)
 VALUES ('Rohan Shah','Pune','rohan@example.com');
INSERT INTO customers (customer_name,city,email)
 VALUES ('Neha Rao','Bengaluru','neha@example.com');
COMMIT;
SELECT customer_id, customer_name, city FROM customers
 ORDER BY customer_id;
```

Expected: 3 rows returned. The COMMIT makes the inserts durable. Each identity value is assigned by Oracle.

USE CASE 04 | Grant a second user read-only access

Let REPORT_USER query APP_USER.CUSTOMERS without changing data. **CHANGES ACCESS**

1. Administrator: create the reporting account

Reconnect locally as SYSDBA, switch to ORCLPDB1, and create only if the account does not exist.

```
CONNECT / AS SYSDBA
ALTER SESSION SET CONTAINER = ORCLPDB1;
SELECT username FROM dba_users WHERE username = 'REPORT_USER';
SET VERIFY OFF
ACCEPT report_pw CHAR PROMPT 'New REPORT_USER password: ' HIDE
CREATE USER REPORT_USER IDENTIFIED BY "&report_pw"
  DEFAULT TABLESPACE USERS ACCOUNT UNLOCK;
UNDEFINE report_pw
GRANT CREATE SESSION TO REPORT_USER;
```

2. Table owner: grant access

```
CONNECT APP_USER@"127.0.0.1:1521/orclpdb1"
GRANT SELECT ON customers TO REPORT_USER;
```

3. Reporting user: verify read access

```
CONNECT REPORT_USER@"127.0.0.1:1521/orclpdb1"
SELECT customer_id, customer_name FROM APP_USER.customers;
-- Optional permission test: INSERT should be denied.
-- Do not grant UPDATE/DELETE or the DBA role.
```

Expected: the SELECT returns three rows. A user does not need CREATE TABLE or DBA privileges merely to query an explicitly granted table.

USE CASE 05 | Monitor storage and sessions

Read basic operational information without modifying the instance. **READ ONLY**

1. Tablespace size and growth

Connect as SYSDBA in ORCLPDB1.

```
SELECT tablespace_name, file_name,
 ROUND(bytes/1024/1024) allocated_mb,
 autoextensible, ROUND(maxbytes/1024/1024) max_mb
FROM dba_data_files
WHERE tablespace_name = 'APP_DATA';
```

2. Current user sessions

```
SELECT username, status, machine, program
FROM v$session
WHERE type = 'USER' AND username IS NOT NULL
ORDER BY username;
```

3. Fast Recovery Area (FRA) utilization

Switch back to CDB root to inspect the overall recovery-area figures.

```
ALTER SESSION SET CONTAINER = CDB$ROOT;
SELECT name,
 ROUND(space_limit/1024/1024) limit_mb,
 ROUND(space_used/1024/1024) used_mb,
 ROUND(space_reclaimable/1024/1024) reclaimable_mb
FROM v$recovery_file_dest;
```

ALERT ARCHIVELOG is enabled in your lab. If FRA usage grows close to its configured 12,732 MB limit, investigate backups and retention before changing or deleting any files. Do not delete archived redo logs using File Explorer.

USE CASE 06 | Take and verify an RMAN database backup

Create a physical backup of the CDB including the PDB and archived redo. **CHANGES FILES**

1. Prepare a backup directory

In Administrator PowerShell, create a lab backup directory. Ensure there is adequate free disk space. Prefer another disk or host for real protection.

```
New-Item -ItemType Directory -Force `
  'C:\OracleLabBackups\RMAN'
Get-Volume | Select-Object DriveLetter, `
  FileSystemLabel, SizeRemaining
```

2. Start RMAN in CDB root

```
$env:ORACLE_SID = 'ORCLCDB'
& "$oh\bin\rman.exe" target /
```

The next commands are entered at RMAN>, not SQL> or PowerShell. This is a whole-CDB backup; do not connect RMAN directly to ORCLPDB1 for the archived redo portion.

3. Configure automatic control-file backup and run backup

```
CONFIGURE CONTROLFILE AUTOBACKUP ON;
RUN {
  ALLOCATE CHANNEL c1 DEVICE TYPE DISK
    FORMAT 'C:\OracleLabBackups\RMAN\%d_%T_%U.bkp';
  BACKUP DATABASE PLUS ARCHIVELOG;
  RELEASE CHANNEL c1;
}
```

A control-file autobackup may be written separately (typically to the FRA unless you configure an alternate control-file autobackup location). Preserve both the backup pieces and control-file autobackup.

4. Validate and inventory the backup

```
LIST BACKUP SUMMARY;
RESTORE DATABASE VALIDATE;
LIST BACKUP OF ARCHIVELOG ALL;
```

A successful RESTORE ... VALIDATE checks restorability of database backups without overwriting live datafiles. It is not a substitute for periodic full restoration on a separate lab environment.

BACKUP SAFETY Keeping RMAN backup pieces on the same C: volume as the database is useful only for training. A disk failure could destroy both. For production use, apply an off-host backup policy, retention, encryption, and recovery testing.

USE CASE 07 | Export and import a sample schema

Copy APP_USER objects into a separate APP_COPY schema using Data Pump. **CHANGES DATA**

1. Prepare an operating-system folder

PowerShell (Administrator): create the Data Pump folder and grant the Oracle database Windows service identity appropriate access. Confirm the account name before granting NTFS permissions.

```
New-Item -ItemType Directory -Force `
  'C:\OracleLabBackups\DataPump'
Get-CimInstance Win32_Service `
  -Filter "Name='OracleServiceORCLCDB'" |
  Select-Object Name, StartName
icacls 'C:\OracleLabBackups\DataPump' `
  /grant 'NT SERVICE\OracleServiceORCLCDB:(OI)(CI)M'
```

2. Create the database DIRECTORY object

In SQL*Plus connected as SYSDBA to ORCLPDB1:

```
ALTER SESSION SET CONTAINER = ORCLPDB1;
CREATE OR REPLACE DIRECTORY LAB_DP_DIR
  AS 'C:\OracleLabBackups\DataPump';
GRANT READ, WRITE ON DIRECTORY LAB_DP_DIR TO APP_USER;
GRANT READ, WRITE ON DIRECTORY LAB_DP_DIR TO SYSTEM;
```

The DIRECTORY is an Oracle object pointing to a server-side Windows folder; it is not an ordinary client file path.

3. Export the APP_USER schema

At PowerShell, exit SQL*Plus and run expdp. Enter APP_USER password only when prompted.

```
& "$oh\bin\expdp.exe" `
 'APP_USER@127.0.0.1:1521/orclpdb1' `
 SCHEMAS=APP_USER DIRECTORY=LAB_DP_DIR `
 DUMPFILE=app_user_lab.dmp LOGFILE=app_user_export.log
```

Check the export log and confirm the .dmp file exists. Avoid using SYSDBA for Data Pump export. If a password contains @, quote the password at the credential prompt (not in the command line), as required by your SQL*Plus lab behavior.

4. Create an empty target user for the import

Connect as SYSDBA to ORCLPDB1; skip creation if APP_COPY already exists.

```
SET VERIFY OFF
ACCEPT copy_pw CHAR PROMPT 'New APP_COPY password: ' HIDE
CREATE USER APP_COPY IDENTIFIED BY "&copy_pw"
  DEFAULT TABLESPACE APP_DATA
  TEMPORARY TABLESPACE TEMP
  QUOTA 200M ON APP_DATA ACCOUNT UNLOCK;
UNDEFINE copy_pw
GRANT CREATE SESSION TO APP_COPY;
```

5. Import under APP_COPY and verify

Run impdp in PowerShell using SYSTEM credentials prompted interactively:

```
& "$oh\bin\impdp.exe" `
 'SYSTEM@127.0.0.1:1521/orclpdb1' `
 DIRECTORY=LAB_DP_DIR DUMPFILE=app_user_lab.dmp `
 LOGFILE=app_copy_import.log `
 REMAP_SCHEMA=APP_USER:APP_COPY
```

If import reports a warning that the pre-created target user already exists, review the entire import log and confirm that its table and data imported. Other errors must be investigated rather than ignored.

```
-- SQL*Plus as SYSDBA in ORCLPDB1:
SELECT owner, table_name FROM dba_tables
 WHERE owner = 'APP_COPY';
SELECT COUNT(*) AS row_count FROM APP_COPY.customers;
```

Expected: APP_COPY.CUSTOMERS exists with 3 rows. A schema export is a logical object/data transfer, not a replacement for RMAN physical backup.

Additional hands-on administration scenarios

The following use cases extend the original guide. They focus on local users in ORCLPDB1, controlled privileges, additional Data Pump modes, and installing the Windows Oracle Client on a separate computer.

SCOPE These are proposed lab exercises and have not been run on your server. Review the affected user and schema before applying any change.

USE CASE 08 | Inspect and manage user accounts

List accounts, unlock an account, reset credentials, and expire passwords. **CHANGES ACCESS**

1. Inventory local PDB users (read-only)

From SQL*Plus as SYSDBA, switch to ORCLPDB1. A user and its schema refer to the same Oracle principal. In the PDB, application users should be local rather than common users.

```
ALTER SESSION SET CONTAINER = ORCLPDB1;
COLUMN username FORMAT A24
COLUMN account_status FORMAT A22
SELECT username, account_status, profile,
      default_tablespace, temporary_tablespace
FROM dba_users
WHERE oracle_maintained = 'N'
ORDER BY username;
```

2. Create a small login-only user

The sample LAB_OPERATOR user intentionally receives only CREATE SESSION; it cannot create tables or query APP_USER tables without further privileges.

```
SELECT username FROM dba_users
WHERE username = 'LAB_OPERATOR';
SET VERIFY OFF
ACCEPT operator_pw CHAR PROMPT 'LAB_OPERATOR password: ' HIDE
CREATE USER LAB_OPERATOR IDENTIFIED BY "&operator_pw"
  DEFAULT TABLESPACE USERS ACCOUNT UNLOCK;
UNDEFINE operator_pw
GRANT CREATE SESSION TO LAB_OPERATOR;
```

IF ALREADY EXISTS If the SELECT returns LAB_OPERATOR, do not repeat CREATE USER. Inspect the existing account and grants first.

3. Manage the account lifecycle

```
-- Account lock blocks new logins:
ALTER USER LAB_OPERATOR ACCOUNT LOCK;
-- Restore logins when authorized:
ALTER USER LAB_OPERATOR ACCOUNT UNLOCK;
-- Require a password update on next login:
ALTER USER LAB_OPERATOR PASSWORD EXPIRE;
-- Review result:
SELECT username, account_status, lock_date, expiry_date
FROM dba_users WHERE username = 'LAB_OPERATOR';
```

4. Reset a password safely

Connect as SYSDBA in ORCLPDB1 and use a hidden SQL*Plus prompt. Avoid writing a real password into shared documents, command history or screenshots.

```
SET VERIFY OFF
ACCEPT reset_pw CHAR PROMPT 'New password: ' HIDE
ALTER USER LAB_OPERATOR IDENTIFIED BY "&reset_pw";
UNDEFINE reset_pw
ALTER USER LAB_OPERATOR ACCOUNT UNLOCK;
```

PASSWORDS If a password contains @, use double quotation marks when SQL*Plus prompts for it during client connections, as established in this lab. Store passwords in an approved password manager.

USE CASE 09 | Grant, audit and revoke privileges

Give users only the rights they need; verify and reverse grants. **CHANGES ACCESS**

1. Understand the three common grant categories

CATEGORY	EXAMPLE	SCOPE
System privilege	CREATE SESSION	Allows login to a PDB
Object privilege	SELECT ON APP_USER.CUSTOMERS	Allows access to one table
Role	LAB_CUSTOMERS_READ	Bundle of privileges granted to users

2. Create a role for read-only reporting

Execute as SYSDBA in ORCLPDB1. APP_USER.CUSTOMERS must already exist (Use Case 03).

```
ALTER SESSION SET CONTAINER = ORCLPDB1;
CREATE ROLE LAB_CUSTOMERS_READ;
GRANT SELECT ON APP_USER.CUSTOMERS TO LAB_CUSTOMERS_READ;
GRANT LAB_CUSTOMERS_READ TO REPORT_USER;
```

Only run the final grant if REPORT_USER has already been created. A role is preferable when multiple users require the same object access.

3. Verify a user's grants

```
SELECT privilege FROM dba_sys_privs
 WHERE grantee = 'REPORT_USER';
SELECT granted_role FROM dba_role_privs
 WHERE grantee = 'REPORT_USER';
SELECT owner, table_name, privilege FROM dba_tab_privs
 WHERE grantee = 'LAB_CUSTOMERS_READ';
```

4. Revoke a privilege and restore it when needed

```
-- Effect: REPORT_USER loses the role's SELECT grant.
REVOKE LAB_CUSTOMERS_READ FROM REPORT_USER;
-- Restore after the access test:
GRANT LAB_CUSTOMERS_READ TO REPORT_USER;
```

CAUTION If REPORT_USER was already granted SELECT directly by APP_USER in Use Case 04, removing a role alone will not remove the direct SELECT grant. Inspect DBA_TAB_PRIVS for REPORT_USER before testing a denied query.

5. Review and adjust storage quotas

```
SELECT tablespace_name, max_bytes
FROM dba_ts_quotas WHERE username = 'APP_USER';
-- Example: limit APP_USER in APP_DATA to 300 MB.
ALTER USER APP_USER QUOTA 300M ON APP_DATA;
```

Apply quota changes only where storage has been approved. A quota is not the same as the tablespace datafile maximum.

USE CASE 10 | Account governance and safe removal

Review inactive accounts and demonstrate a cautious cleanup procedure. **CHANGES ACCESS**

1. Check privileges and active sessions

```
SELECT username, account_status, created
FROM dba_users WHERE username = 'LAB_OPERATOR';
SELECT sid, serial#, username, status, machine
FROM v$session WHERE username = 'LAB_OPERATOR';
```

2. Retire a login without deleting its objects

```
ALTER USER LAB_OPERATOR ACCOUNT LOCK;
REVOKE CREATE SESSION FROM LAB_OPERATOR;
```

This is safer than immediately dropping an account. Verify whether applications or database jobs depend on the account before removing it.

3. Optional lab-only cleanup

```
-- Execute only for your disposable LAB_OPERATOR account
-- after checking that no objects/data must be retained:
SELECT object_name, object_type FROM dba_objects
 WHERE owner = 'LAB_OPERATOR';
-- If empty and removal is approved:
DROP USER LAB_OPERATOR;
```

DESTRUCTIVE COMMAND Do not use DROP USER ... CASCADE casually: it permanently removes the user's owned database objects. No DROP USER execution is required for the learning objectives.

USE CASE 11 | Additional Data Pump export/import modes

Learn single-table, metadata-only, and repeatable export techniques. **CHANGES FILES**

Prerequisites: complete Use Cases 03 and 07. The server-side folder C:\OracleLabBackups\DataPump and Oracle directory LAB_DP_DIR must exist and be accessible to the database service account.

1. Export only one table

Run in PowerShell; the export produces a dump and a log on the DATABASE SERVER, even if expdp runs from an installed remote client.

```
& "$oh\bin\expdp.exe" `
 'APP_USER@127.0.0.1:1521/orclpdb1' `
 TABLES=CUSTOMERS DIRECTORY=LAB_DP_DIR `
 DUMPFILE=customers_only.dmp `
 LOGFILE=customers_only_export.log
```

2. Export schema structure, without table rows

```
& "$oh\bin\expdp.exe" `
 'APP_USER@127.0.0.1:1521/orclpdb1' `
 SCHEMAS=APP_USER CONTENT=METADATA_ONLY `
 DIRECTORY=LAB_DP_DIR DUMPFILE=app_metadata.dmp `
 LOGFILE=app_metadata_export.log
```

CONTENT=METADATA_ONLY captures object definitions, not the CUSTOMERS row data. Use it to review a schema structure migration rather than a full data migration.

3. Use a parameter file for repeatable exports

On the CLIENT machine, create C:\OracleLabBackups\app_export.par with these lines (no password):

```
SCHEMAS=APP_USER
DIRECTORY=LAB_DP_DIR
DUMPFILE=app_user_repeat.dmp
LOGFILE=app_user_repeat.log
```

Then run in PowerShell:

```
& "$oh\bin\expdp.exe" `
 'APP_USER@127.0.0.1:1521/orclpdb1' `
 PARFILE='C:\OracleLabBackups\app_export.par'
```

4. Verify exports on the database server

```
Get-ChildItem 'C:\OracleLabBackups\DataPump' |
 Select-Object Name, Length, LastWriteTime
Get-Content 'C:\OracleLabBackups\DataPump\app_user_repeat.log' `
 -Tail 25
```

DATA PUMP FILE LOCATIONS A Data Pump parameter file resides on the initiating client, but dump/log files referenced through DIRECTORY are written/read on the database server. Copy a .dmp to another database server before an offline import.

USE CASE 12 | Import schema into a different user

Remap APP_USER objects to APP_COPY and validate imported data. **CHANGES DATA**

1. Confirm the destination user

The original guide creates APP_COPY in ORCLPDB1 with a quota on APP_DATA. Verify that account exists, is unlocked, and the original APP_USER export completed successfully.

```
-- SQL*Plus, SYSDBA, ORCLPDB1
SELECT username, account_status, default_tablespace
FROM dba_users WHERE username = 'APP_COPY';
SELECT tablespace_name, max_bytes
FROM dba_ts_quotas WHERE username = 'APP_COPY';
```

2. Import objects into APP_COPY

PowerShell; use the SYSTEM password when prompted. SYSTEM is a privileged import account; use it only for this controlled administrative operation.

```
& "$oh\bin\impdp.exe" `
 'SYSTEM@127.0.0.1:1521/orclpdb1' `
 DIRECTORY=LAB_DP_DIR DUMPFILE=app_user_lab.dmp `
 LOGFILE=app_copy_import.log `
 REMAP_SCHEMA=APP_USER:APP_COPY
```

3. Verify objects and counts

```
SELECT owner, table_name FROM dba_tables
 WHERE owner = 'APP_COPY';
SELECT COUNT(*) AS row_count FROM APP_COPY.CUSTOMERS;
```

Expected: CUSTOMERS exists, with the same 3 sample rows. If re-running the import into an existing schema, first inspect the logfile: pre-existing tables may be skipped. Do not apply TABLE_EXISTS_ACTION=REPLACE without an explicit change plan.

4. Moving a dump to a second Oracle server

Export on the source, copy the dump file securely to a folder accessible to the target Oracle database service, create a DIRECTORY object on the target PDB, check target version and privileges, then run impdp against the target service. A client installation alone cannot provide the server-side dump-file folder.

BACKUP DISTINCTION Data Pump is logical object/data transfer, not a substitute for RMAN physical database backup and recoverability tests.

USE CASE 13 | Install Oracle Database 19c Client on Windows

Install client tools on another Windows machine for remote administration. **CHANGES OS**

1. Determine whether a separate client is needed

Your Windows database server already has SQL*Plus, expdp, impdp and Oracle Net binaries. Install a separate Oracle Client only on a workstation or application host needing Oracle client tools. The standalone Client does not install a database engine.

CLIENT TYPE	WHEN TO CHOOSE
Administrator	SQL*Plus and administration / Data Pump utilities
Runtime	Applications needing Oracle networking and client libraries
Instant Client	Smaller deployment when only necessary client libraries/tools are wanted

2. Choose the appropriate installation archive

Your previously downloaded V982658-01.zip was listed as a Windows x64 Oracle Database Client package. Check its actual extracted contents: Oracle distributes BOTH traditional client installation media and a client_home (gold image) ZIP. Their extraction methods differ.

```
# PowerShell - locate the downloaded archive
Get-Item "$env:USERPROFILE\Downloads\V982658-01.zip"
# Extract into a temporary staging folder if using traditional media:
New-Item -ItemType Directory -Force 'C:\OracleInstall\client19c'
Expand-Archive -LiteralPath `
 "$env:USERPROFILE\Downloads\V982658-01.zip" `
 -DestinationPath 'C:\OracleInstall\client19c'
Get-ChildItem 'C:\OracleInstall\client19c' `
 -Recurse -Filter setup.exe | Select-Object FullName
```

3A. Traditional installation media

If the archive contains a media directory with setup.exe (for example under a client folder), start that setup.exe as Administrator from its extracted media location. Do not move traditional-media installation contents into an Oracle Home merely because the server used an image-based installer.

3B. Client gold image (client_home)

If the extracted archive is the client_home image, extract its contents directly into a dedicated client Oracle Home (not the existing database home), then run setup.exe from that home. Oracle documents this as the image-based installation approach.

```
# Example target on a separate client computer:
C:\app\oracleclient\product\19.0.0\client_1
```

4. Recommended GUI wizard choices

WIZARD FIELD	RECOMMENDATION
Installation type	Administrator for this admin-learning use case
Oracle Base	C:\app\oracleclient
Client Home	C:\app\oracleclient\product\19.0.0\client_1
Prerequisite checks	Do not bypass OS failures; install a supported RU
Other Oracle homes	Keep client and database homes separate

OS CERTIFICATION Oracle Database 19.3 (server) is not certified for Windows Server 2025; its server requires at least RU 19.27. For the separate client machine, confirm client OS certification and the client software update level before installation. Do not interpret an installer passing checks as certification.

5. Configure a client TNS alias

Create/edit tnsnames.ora under the CLIENT Oracle Home network\admin folder, or configure the same alias using Net Configuration Assistant. Example:

```
ORCLPDB1_LAB =
 (DESCRIPTION =
   (ADDRESS = (PROTOCOL = TCP)
    (HOST = 192.168.122.120)(PORT = 1521))
   (CONNECT_DATA = (SERVICE_NAME = orclpdb1))
 )
```

The recorded server IP 192.168.122.120 is from your NAT-style VM network. A separate PC may not have a route to that private subnet; verify reachability and choose an authorized reachable IP or routed/bridged configuration. Restrict port 1521 to trusted clients.

6. Test from the Windows client

```
$clientHome = 'C:\app\oracleclient\product\19.0.0\client_1'
Test-Path "$clientHome\bin\sqlplus.exe"
Test-NetConnection 192.168.122.120 -Port 1521
& "$clientHome\bin\tnsping.exe" ORCLPDB1_LAB
& "$clientHome\bin\sqlplus.exe" /nolog
```

At the SQL> prompt (not PowerShell):

```
CONNECT REPORT_USER@ORCLPDB1_LAB
SHOW USER;
SHOW CON_NAME;
SELECT COUNT(*) FROM APP_USER.CUSTOMERS;
```

Enter the reporting-user password when prompted. For the direct schema connection use REPORT_USER; for administrative examples use SYSTEM but avoid using SYSDBA over a remotely exposed network.

7. Resolve common client problems

SYMPTOM	CHECK
SQL*Plus not found	Use explicit <clientHome>\bin\sqlplus.exe; inspect PATH / where.exe sqlplus
ORA-12154	Alias spelling, tnsnames.ora location, TNS_ADMIN, naming methods
ORA-12514	Service orclpdb1 is registered and READY on listener
ORA-12543	Compare sqlnet.log descriptor; connection syntax and @ password quoting
TCP failure	VM routing, Windows firewall, listener port, target address

Daily Oracle administrator cheat sheet

OBJECTIVE	COMMAND / LOCATION
Database and logging	SQL> SELECT name, open_mode, log_mode FROM v$database;
Current PDB	SQL> SHOW CON_NAME;
PDB inventory	SQL> SHOW PDBS;
Current session	SQL> SHOW USER;
Tablespaces	SQL> SELECT tablespace_name FROM dba_tablespaces;
Account status	SQL> SELECT username, account_status FROM dba_users;
Listener	PS> & "$oh\bin\lsnrctl.exe" status LISTENER
Windows services	PS> Get-Service *Oracle*
Port 1521	PS> Test-NetConnection 127.0.0.1 -Port 1521
Recent alert log	Oracle ADR / diagnostic files under C:\app\oracle\diag
RMAN inventory	RMAN> LIST BACKUP SUMMARY;
Data Pump output	Review .log in C:\OracleLabBackups\DataPump

Typical beginner errors and interpretation

ERROR	FIRST CHECK
SP2-0306 / invalid option	SQL*Plus CONNECT quoting and whether password contains @.
ORA-12543	Inspect client sqlnet.log; compare HOST and SERVICE_NAME in failed descriptor.
ORA-12514	Verify listener services and PDB service registration.
ORA-01017	Credentials, exact user, quoting and target PDB.
ORA-65096	You are attempting local-user creation in CDB$ROOT; switch to ORCLPDB1.
ORA-01950	User has insufficient quota on the target tablespace.

ORA-39087 / ORA-29283	Data Pump directory object, Windows folder ACLs and service identity.
ORA-19809 / FRA pressure	Check v$recovery_file_dest and RMAN backup/retention status.

Execution checklist

- ☐ Verified SYSDBA connection and PDB open mode
- ☐ Checked Oracle listener and Windows services
- ☐ Created APP_USER and populated CUSTOMERS with 3 rows
- ☐ Created REPORT_USER and verified read-only access
- ☐ Reviewed tablespace and FRA metrics
- ☐ Completed RMAN database-plus-archivelog backup and validation
- ☐ Exported APP_USER and imported as APP_COPY; verified row count

Official Oracle references

Windows installation requirements | https://docs.oracle.com/en/database/oracle/oracle-database/19/ntdbi/operating-system-checklist-oracle-database-installation-microsoft-windows.html

Backup and Recovery User Guide | https://docs.oracle.com/en/database/oracle/oracle-database/19/bradv/

RMAN backup procedure | https://docs.oracle.com/en/database/oracle/oracle-database/19/bradv/getting-started-rman.html

Oracle Data Pump Export | https://docs.oracle.com/en/database/oracle/oracle-database/19/sutil/oracle-data-pump-export-utility.html

Oracle Data Pump Import | https://docs.oracle.com/en/database/oracle/oracle-database/19/sutil/datapump-import-utility.html

Oracle SQL GRANT syntax | https://docs.oracle.com/en/database/oracle/oracle-database/19/sqlrf/GRANT.html

Oracle ALTER USER | https://docs.oracle.com/en/database/oracle/oracle-database/19/sqlrf/ALTER-USER.html

Windows 19c Client Installation | https://docs.oracle.com/en/database/oracle/oracle-database/19/ntcli/

Client image installation | https://docs.oracle.com/en/database/oracle/oracle-database/19/ntcli/installing-oracle-database-client-using-image-files.html

Oracle Client Windows downloads | https://www.oracle.com/database/technologies/oracle19c-windows-downloads.html

Document status: Training guide. Steps are illustrative and are to be executed/validated individually. This document does not assert that the newly described exercises have been run on the server.